Insurance Claim Analytics

Operational Efficiency & Risk Detection Platform

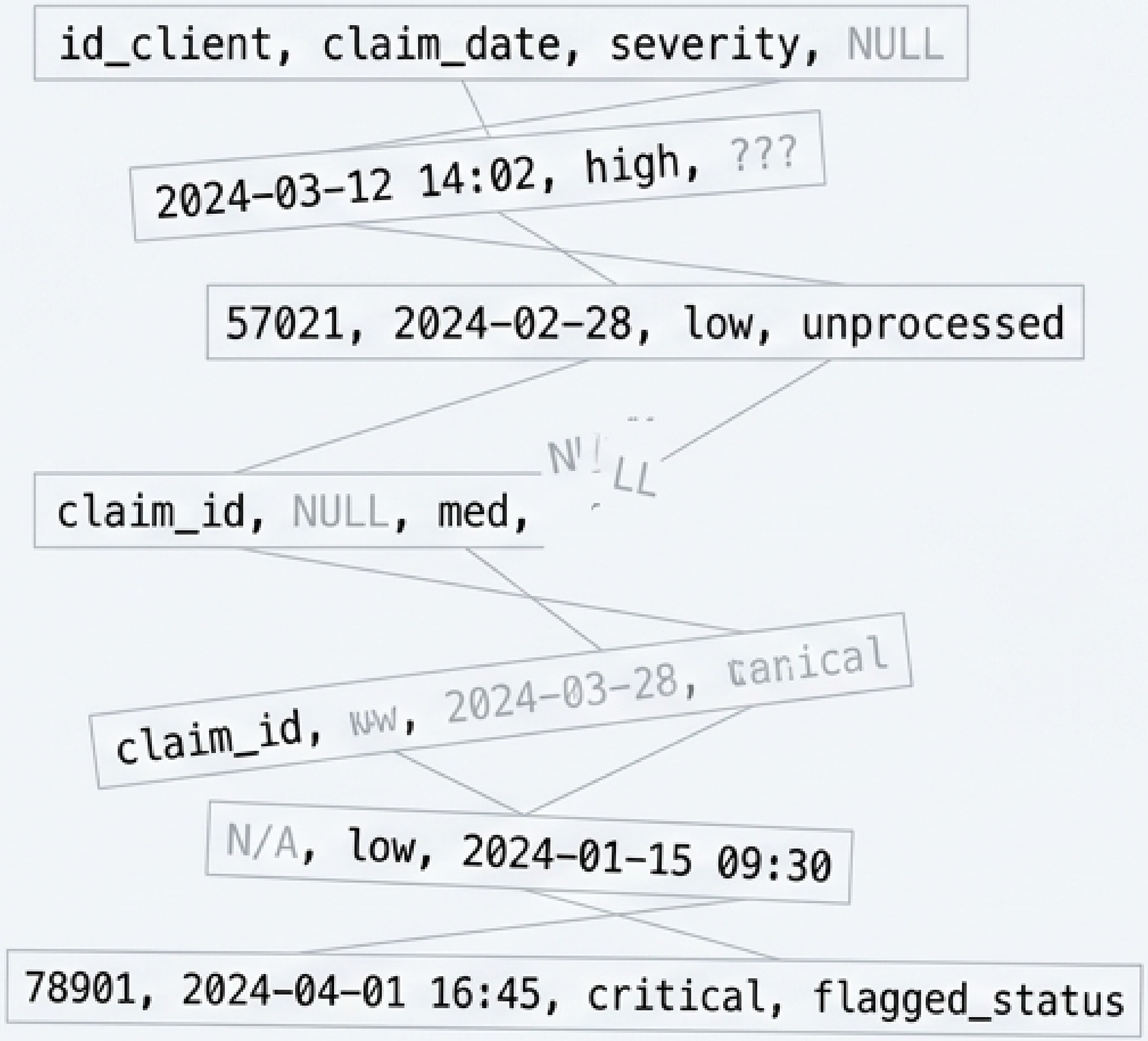
[BUILD] R & Tidyverse

[CORE] Synthetic Data Generation

[OUTPUT] Automated Business Intelligence

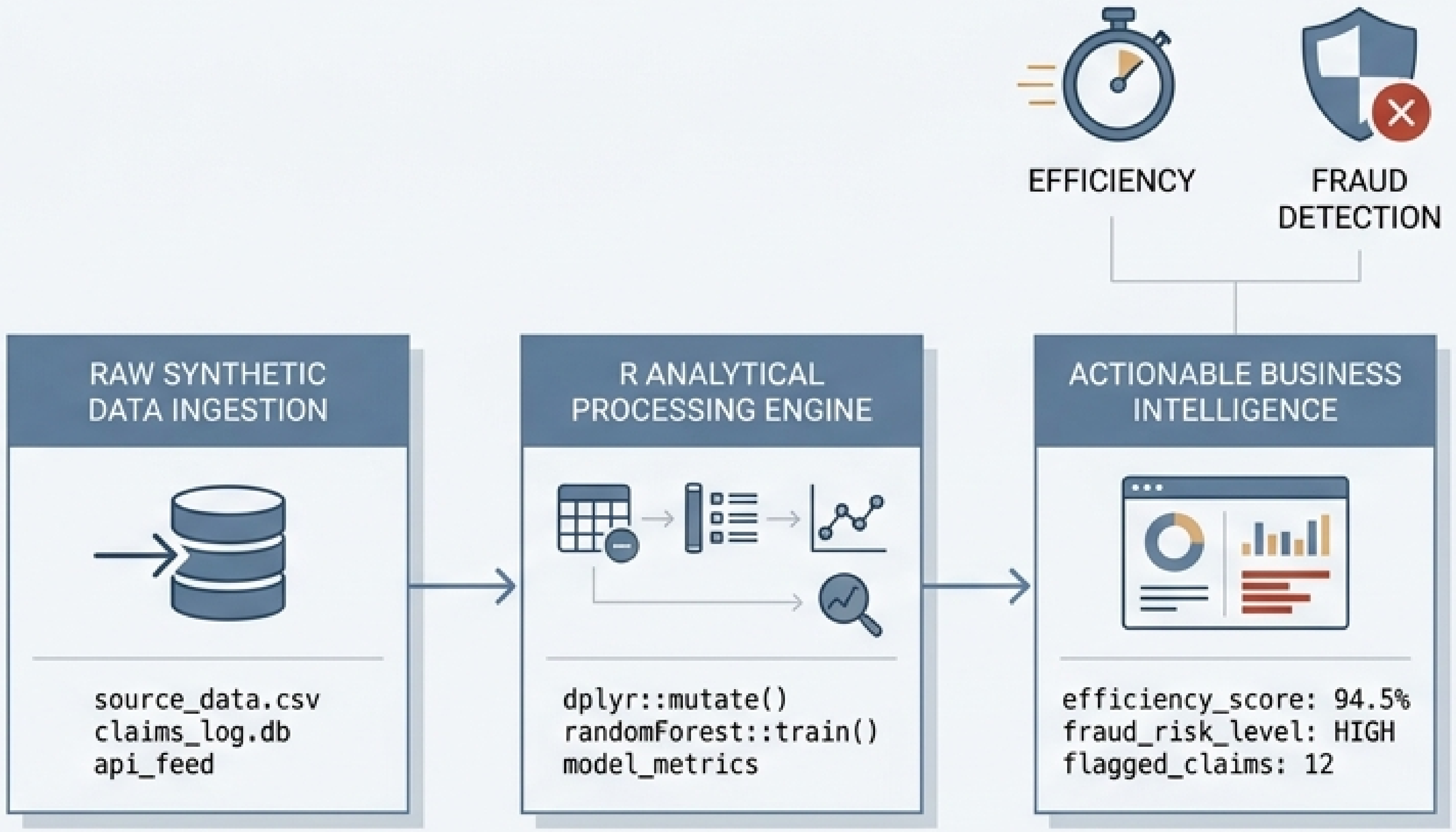
THE OPERATIONAL BLINDSPOT

Fragmented claim data conceals operational bottlenecks and masks concentrated fraudulent behavior.



AUTOMATED INTELLIGENCE PIPELINE

An end-to-end analytical solution built in R. It translates raw, synthetic claims data into actionable business intelligence—measuring processing efficiency and detecting fraudulent patterns simultaneously.



Identified 60-Day Bottleneck

Uncovered severe resolution delays specifically in Health claims, enabling targeted process redesign.

Isolated Concentrated Fraud

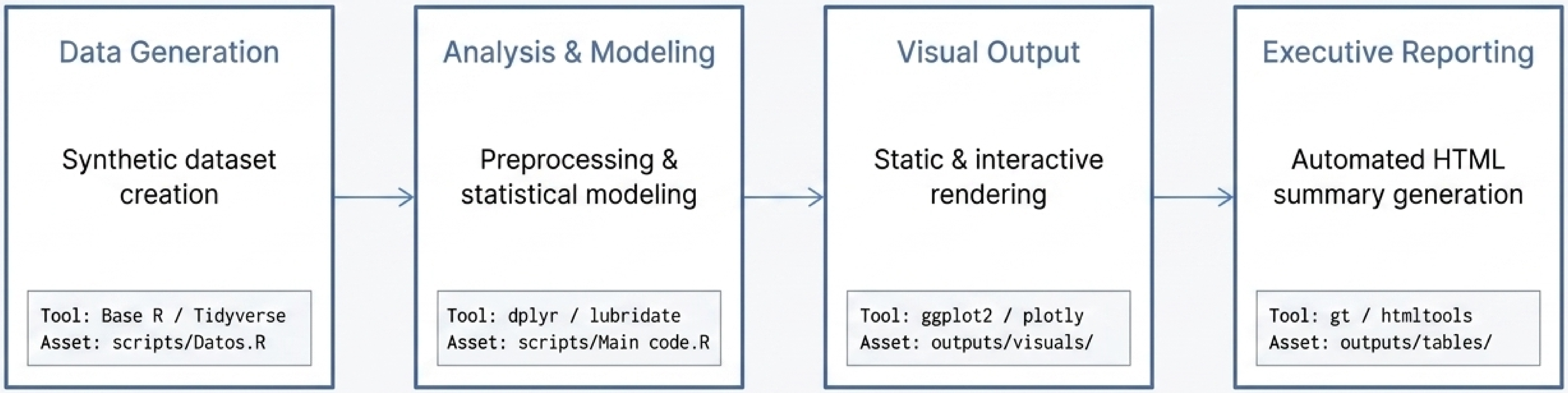
Deployed a multi-variable scoring system to pinpoint high-risk client behaviors and flag anomalies.

Severity as Primary Driver

Statistically validated that claim severity is the dominant, predictable driver of resolution time.

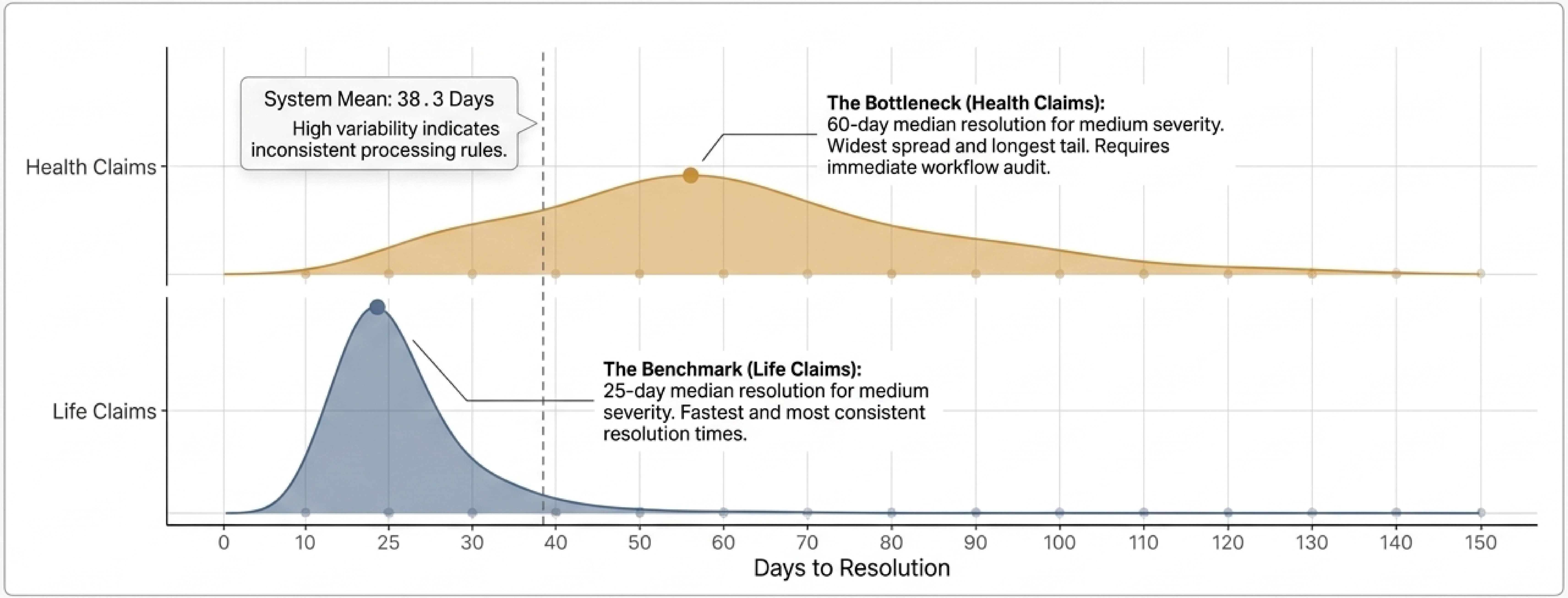
Established Cross-Product Standards

Created baseline processing standards across Life, Auto, Health, and Home insurance portfolios.


Data Generation
Synthetic dataset creation
Tool: Base R / Tidyverse
Asset: scripts/Datos.R
Analysis & Modeling
Preprocessing & statistical modeling
Tool: dplyr / lubridate
Asset: scripts/Main code.R
Visual Output
Static & interactive rendering
Tool: ggplot2 / plotly
Asset: outputs/visuals/
Executive Reporting
Automated HTML summary generation
Tool: gt / htmltools
Asset: outputs/tables/

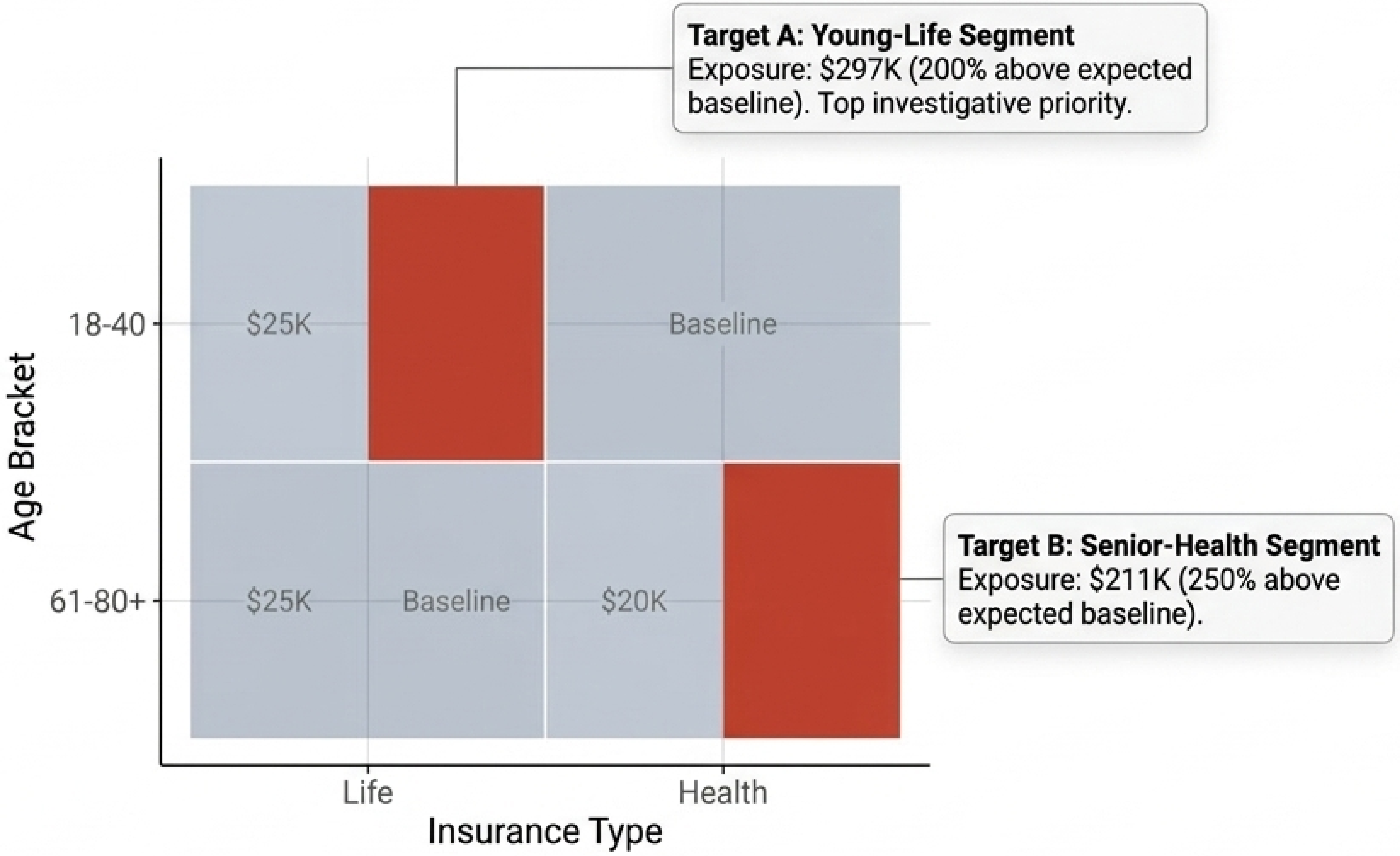
Gemini Notebook

Insight I: Processing Efficiency & Bottlenecks



Key Driver: Clear monotonic relationship identified: as severity increases, time-to-resolution increases predictably across all segments.

Insight II: High-Risk Demographic Segments



System Anomalies Detected

Age Gap: Absolute zero health claims under age 41 despite high frequencies in older cohorts.

Auto Spikes: Highly concentrated claim activity localized in the 51-60 age group.

Life Patterns: Claim frequency exclusively appears within the 61-80 age range.

Insight III: Isolated Fraud Patterns

System Alert

`flag_client_concentration`

CRITICAL PRIORITY: Client Concentration

Client #12 submitted 6 claims within a 30-day window, accounting for 40% of all high-risk flags in the system.

System Alert

`flag_rapid_closure < 24h`

HIGH PRIORITY: Rapid Resolution

Suspicious 1-day claim closures detected, bypassing standard review protocols.

System Alert

`flag_severity_cluster`

MEDIUM PRIORITY: Severity Correlation

70% of all claims flagged as high-risk are also categorized as high severity, indicating targeted large-payout attempts.

The Fraud Risk Scoring Engine

Scoring Diagnostic Table

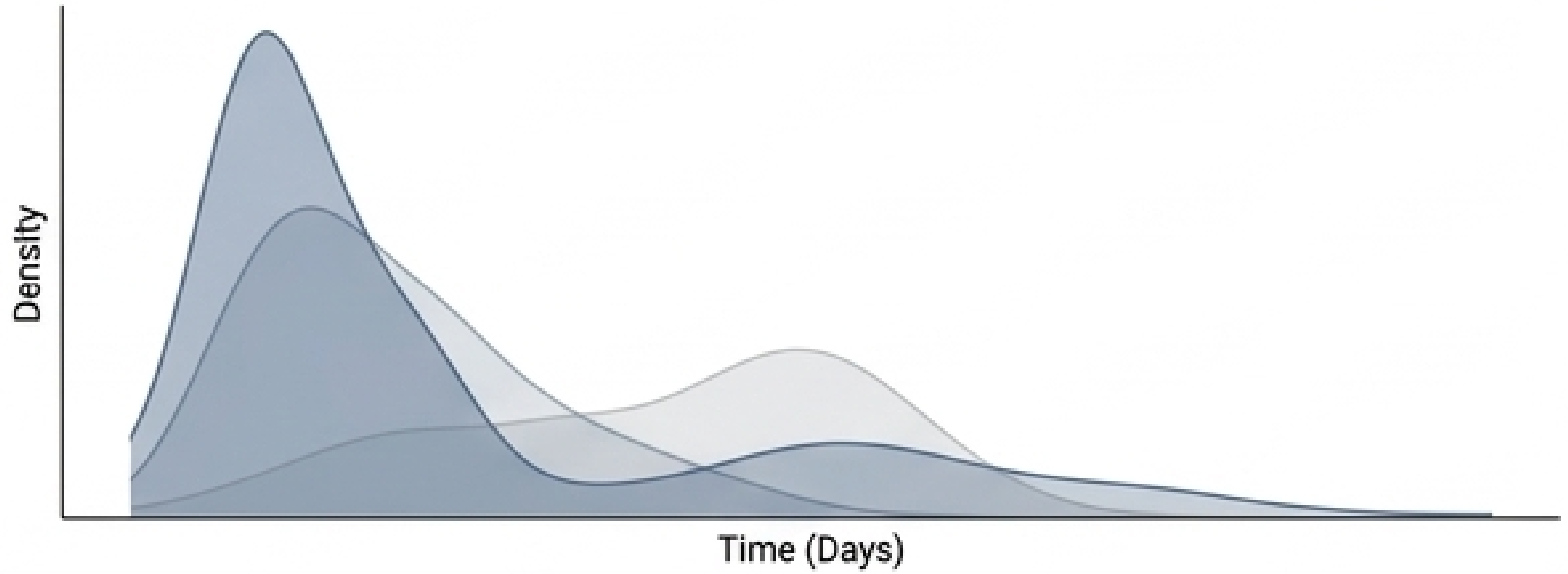
Rapid Resolution	Claims closed in <5 days	**[+25 pts]**
High Frequency	>5 claims per client in 30 days	**[+25 pts]**
Extreme Amounts	Claimed value >95th percentile	**[+20 pts]**
Disparity	Claimed amount >80% of total insured value	**[+20 pts]**
Severity Flags	Claim logged as High Severity	**[+10 pts]**

Risk Classification Tiers

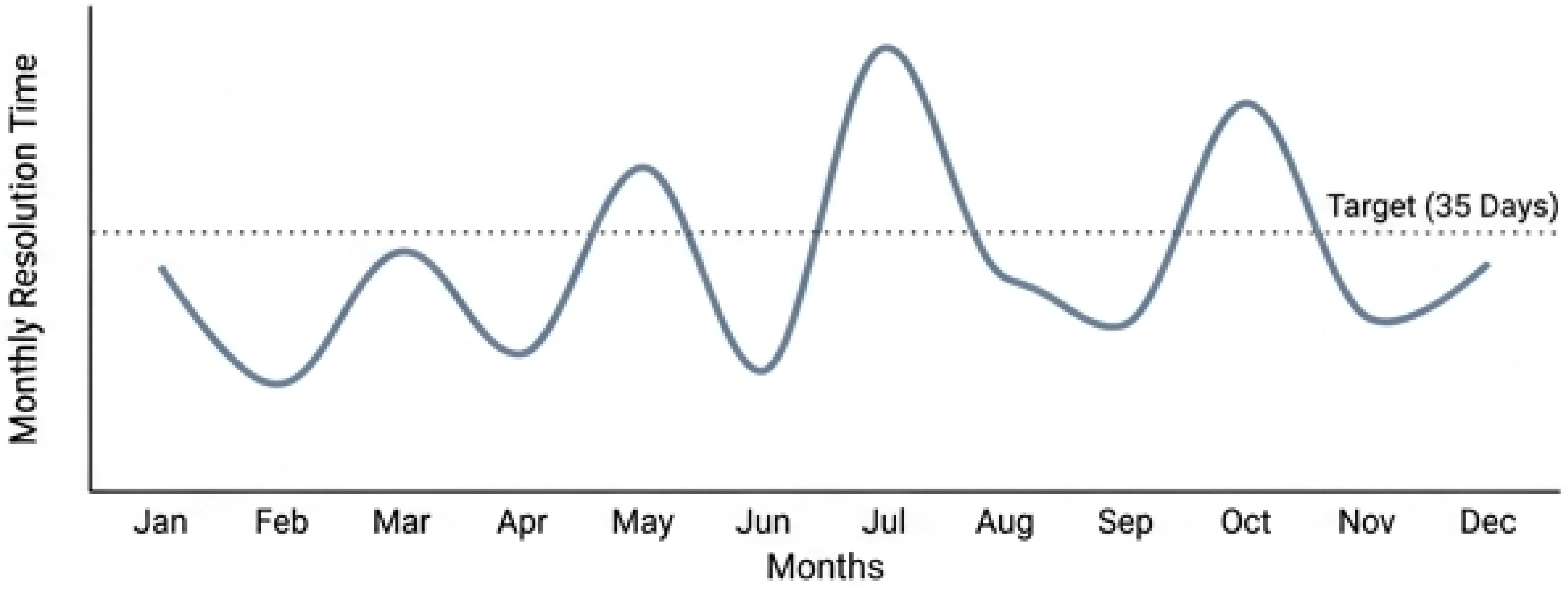
- Critical: >45 Points
- High: 35-44 Points
- Medium: 25-34 Points
- Low: <25 Points

```
System Note: Overall portfolio health remains strong (93% at healthy baseline),
isolating investigative focus on precise targets.
```

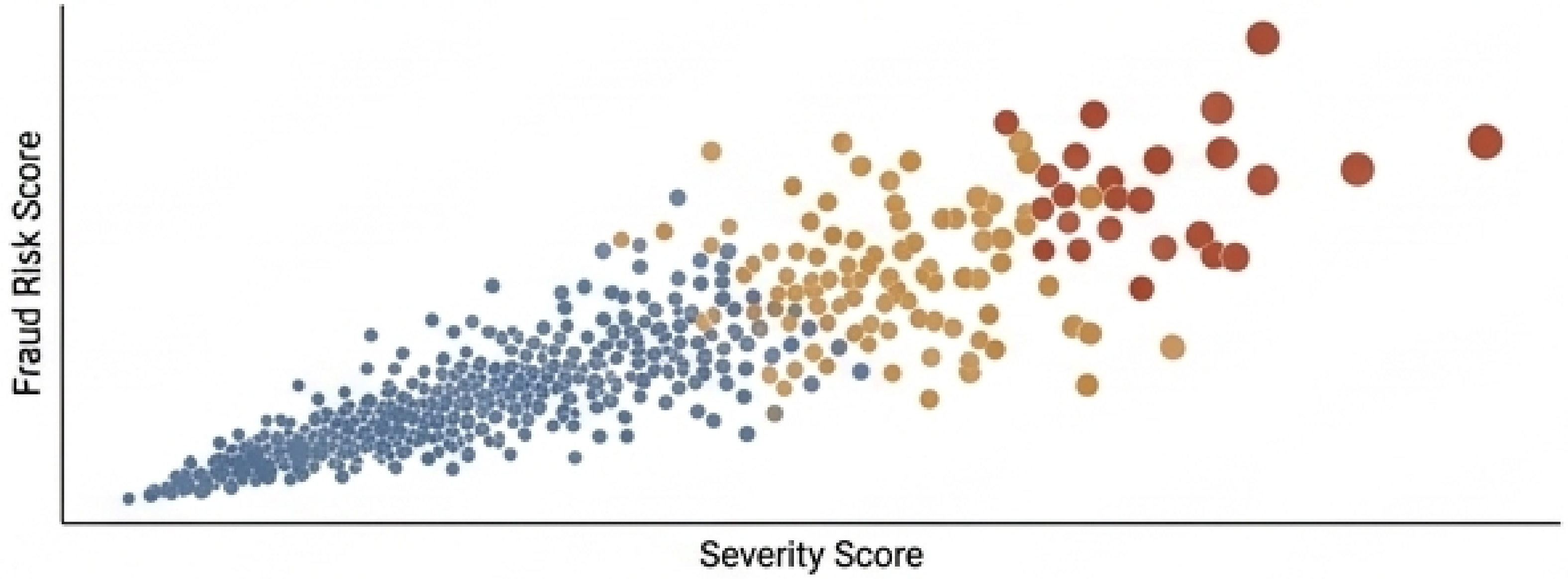
Automated Visual Analytics Suite



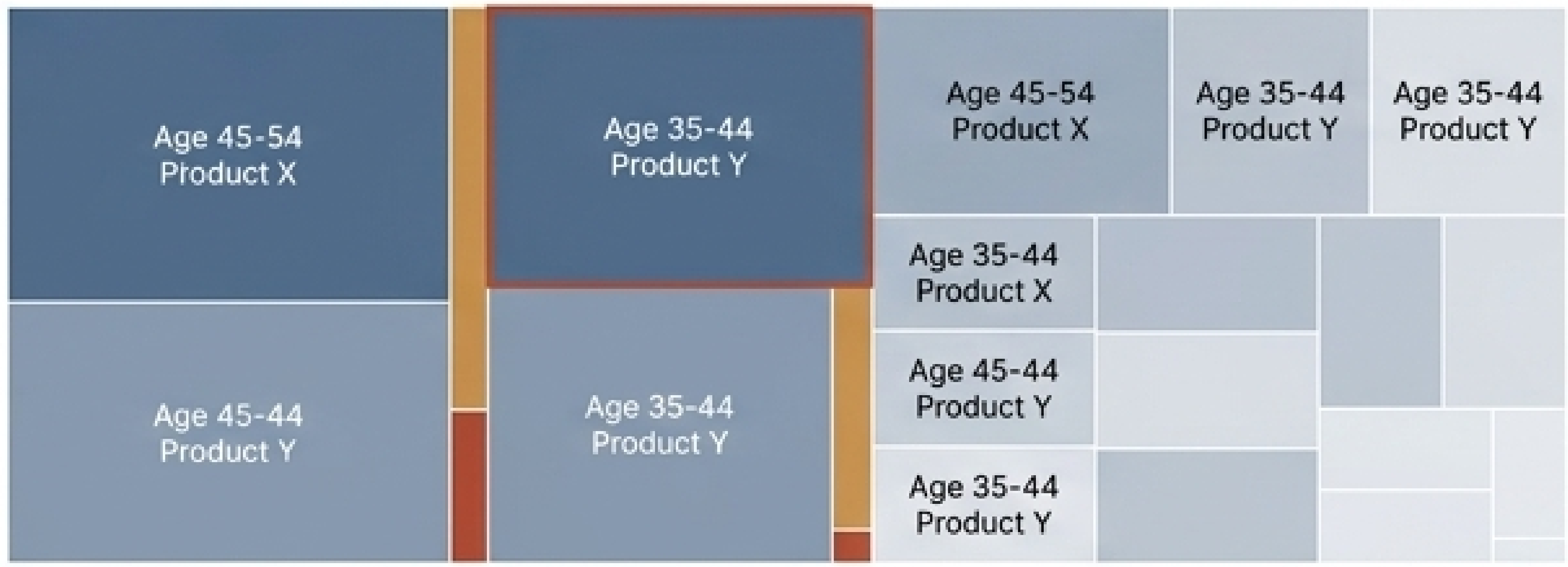
Resolution Distribution: Multi-panel views isolating workflow tails and outlining severity as the primary driver of resolution time.



Monthly Resolution Trend: Smoothed trend lines identifying cyclical bottlenecks exceeding the 35-day target benchmark.

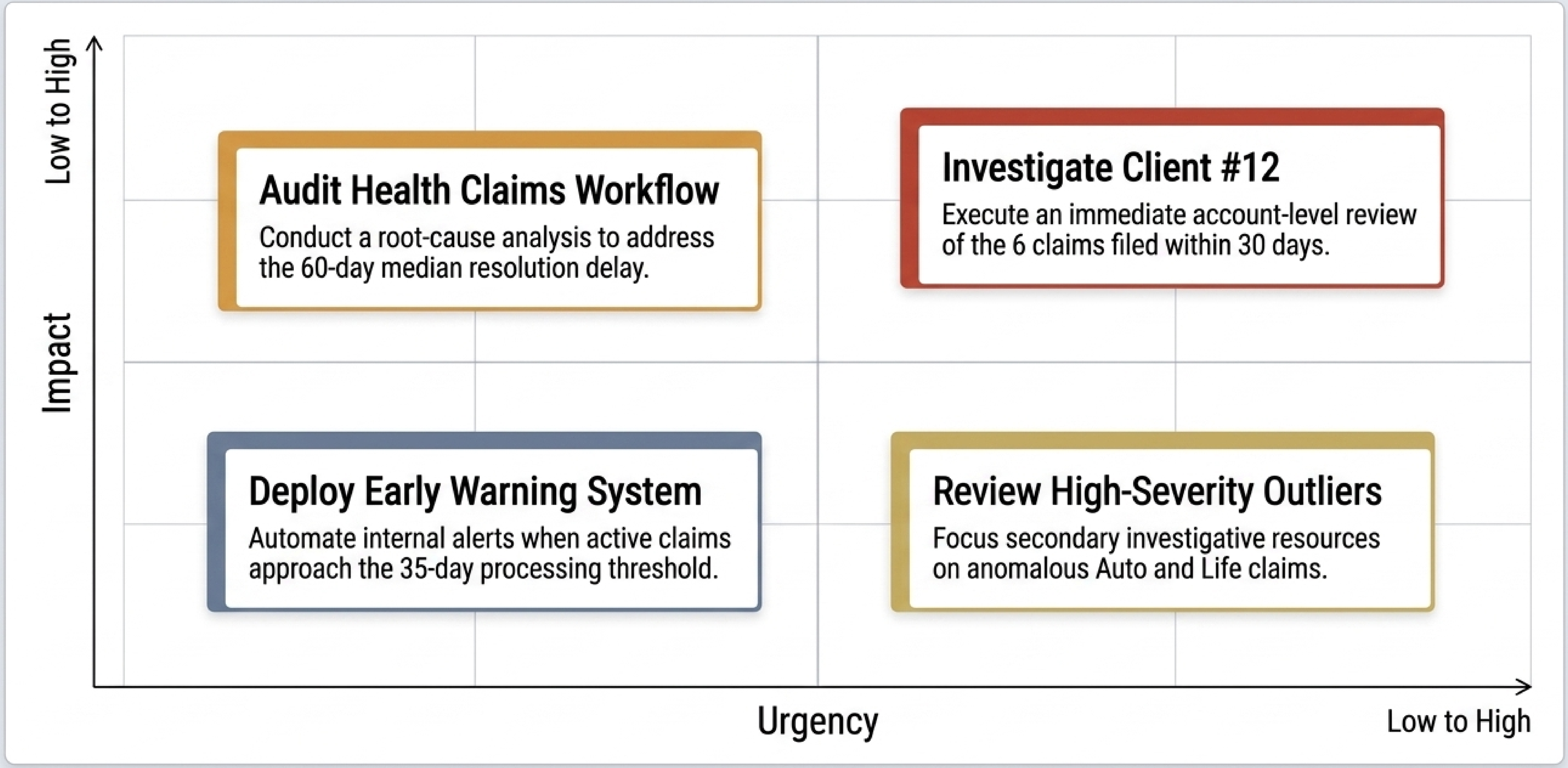


Fraud Risk Matrix: Plotting risk score distribution against severity, visualizing heteroscedasticity across segments.



Demographic Exposure: Highlighting absolute dollar exposure hotspots by age bracket and insurance product.

Strategic Interventions: Immediate Action



Analytics Roadmap: Future Capabilities

Next Quarter

Segment-Based Workflows

Implement dynamic, differentiated processing tracks based automatically on initial claim severity

Next 6 Months

Client-Level Aggregate Scoring

Shift from individual claim scoring to persistent, lifetime fraud risk scores for every client entity

Year 1

Predictive Modeling

Transition from descriptive baselines to ML-driven time-to-resolution forecasting using severity features

Year 1.5

Network Analysis

Deploy graph analytics to uncover hidden, connected fraud rings operating across multiple client IDs

Operationalizing the Platform

STEP 1: INITIAL SETUP

Clone repository and configure local environment path.

```
git clone https://github.com/Michell-Matias-Tello/Insurance-Claim-Analytics.git
```

STEP 2: GENERATE SYNTHETIC DATA

Create base policies, clients, claims, and payment CSVs.

```
source("scripts/Cargar los paquetes.txt")
```

STEP 3: EXECUTE PIPELINE

Run preprocessing, scoring, and automated visual generation.

```
source("scripts/Main code.R")
```

Open Source Intelligence

Author: Michell Matias Tello

License: MIT License (Open for enterprise integration)

Standard: Tidyverse style guide compliant

```
Michell-Matias-Tello/Insurance-Claim-Analytics
```

Status: Completed – Ready for Review

Action: Contributions welcome via Pull Request

Full Documentation: Link to Notion comprehensive analysis overview.

Star this repository if you find it useful.